

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – CONCEPT MAPPING

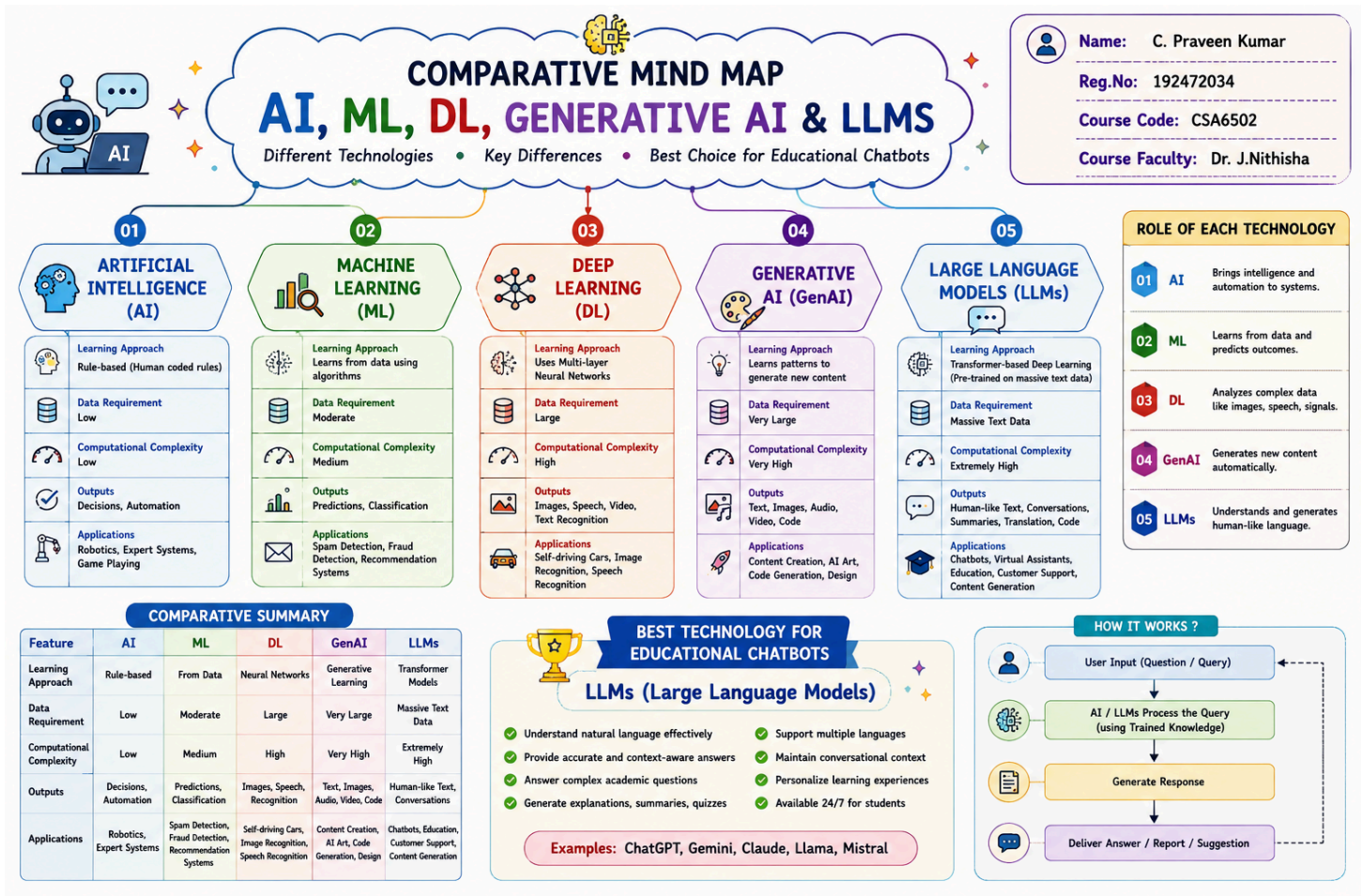
Course Code:	CSA6502	Course Title:	Generative AI and Large Scale Model
CO Assessed:	CO1-Imitation (BL3)	Assessment:	Assessment Tool 1– CONCEPT MAPPING
Weightage:	10% of CIA	Total Marks:	25
CO1 Statement:	Explain the fundamentals of Artificial Intelligence, Generative AI, Foundation Models, Transformer architecture, tokenization, embeddings, and Large Language Models.		
Student Name:	C.Praveen Kumer	Reg. No.:	192472034
Section / Batch:	2024	Date:	15-07-2026

Code of Conduct

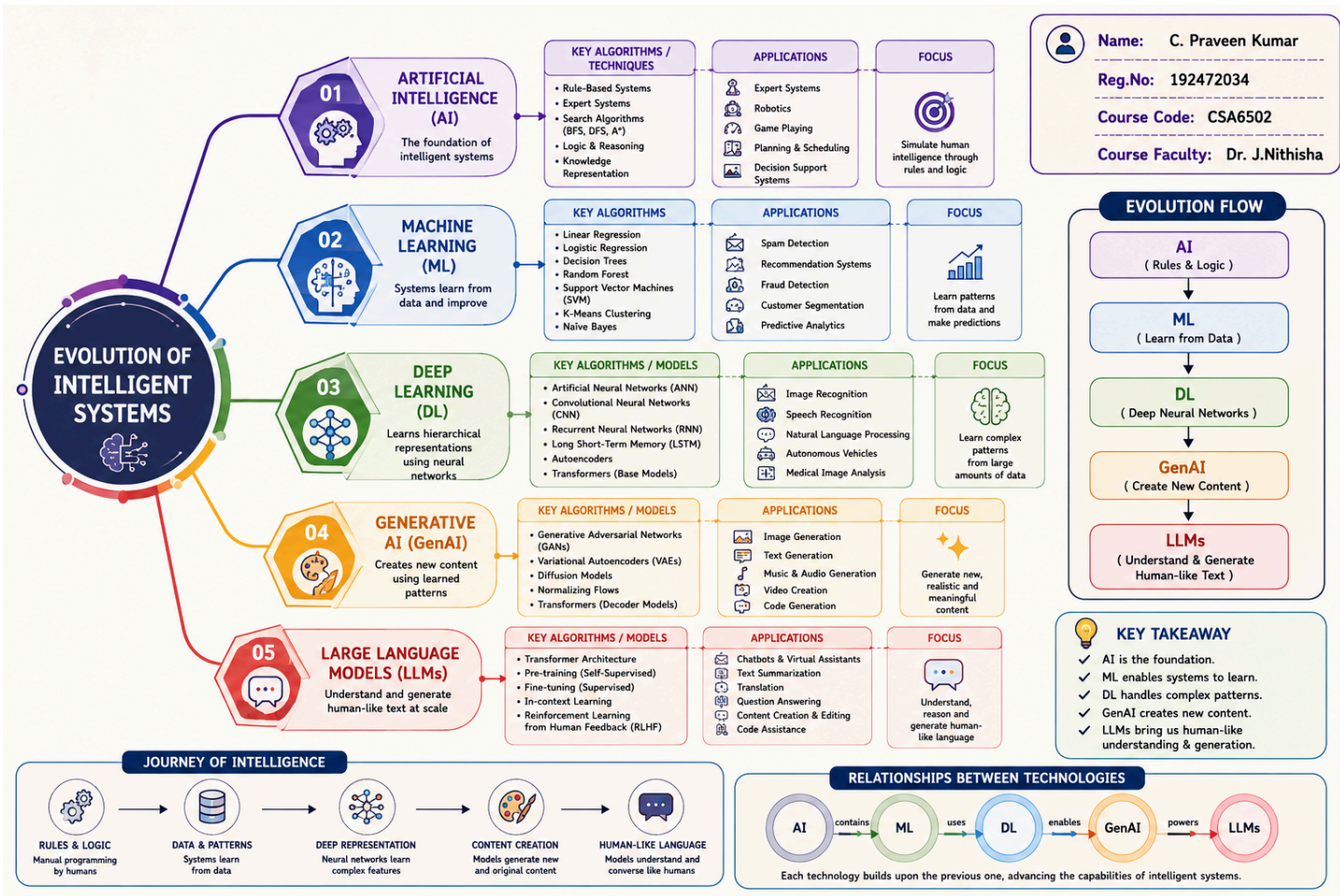
I C.Praveen Kumar (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: C.Praveen kumar

1. Create a comparative mind map illustrating the differences among AI, ML, DL, Generative AI, and LLMs based on learning approach, data requirements, computational complexity, outputs, and applications. Recommend the best technology for educational chatbots.



2. Design a comprehensive mind map for the evolution of intelligent systems, beginning with Artificial Intelligence and progressing through Machine Learning, Deep Learning, Generative AI, and Large Language Models. Include key algorithms, applications, and relationships between each technology.



03

DEEP LEARNING (DL)

Learns hierarchical representations using neural networks

KEY ALGORITHMS / MODELS

- Artificial Neural Networks (ANN)
- Convolutional Neural Networks (CNN)
- Recurrent Neural Networks (RNN)
- Long Short-Term Memory (LSTM)
- Autoencoders
- Transformers (Base Models)

APPLICATIONS

- Image Recognition
- Speech Recognition
- Natural Language Processing
- Autonomous Vehicles
- Medical Image Analysis

FOCUS

- Learn complex patterns from large amounts of data

04

GENERATIVE AI (GenAI)

Creates new content using learned patterns

KEY ALGORITHMS / MODELS

- Generative Adversarial Networks (GANs)
- Variational Autoencoders (VAEs)
- Diffusion Models
- Normalizing Flows
- Transformers (Decoder Models)

APPLICATIONS

- Image Generation
- Text Generation
- Music & Audio Generation
- Video Creation
- Code Generation

FOCUS

- Generate new, realistic and meaningful content

05

LARGE LANGUAGE MODELS (LLMs)

Understand and generate human-like text at scale

KEY ALGORITHMS / MODELS

- Transformer Architecture
- Pre-training (Self-Supervised)
- Fine-tuning (Supervised)
- In-context Learning
- Reinforcement Learning from Human Feedback (RLHF)

APPLICATIONS

- Chatbots & Virtual Assistants
- Text Summarization
- Translation
- Question Answering
- Content Creation & Editing
- Code Assistance

FOCUS

- Understand, reason and generate human-like language

EVOLUTION FLOW

AI
(Rules & Logic)

↓

ML
(Learn from Data)

↓

DL
(Deep Neural Networks)

↓

GenAI
(Create New Content)

↓

LLMs
(Understand & Generate Human-like Text)

KEY TAKEAWAY

- ✓ AI is the foundation.
- ✓ ML enables systems to learn.
- ✓ DL handles complex patterns.
- ✓ GenAI creates new content.
- ✓ LLMs bring us human-like understanding & generation.

JOURNEY OF INTELLIGENCE

RULES & LOGIC
Manual programming by humans

→

DATA & PATTERNS
Systems learn from data

→

DEEP REPRESENTATION
Neural networks learn complex features

→

CONTENT CREATION
Models generate new and original content

→

HUMAN-LIKE LANGUAGE
Models understand and converse like humans

RELATIONSHIPS BETWEEN TECHNOLOGIES

AI

contains

ML

uses

DL

enables

GenAI

powers

LLMs

Each technology builds upon the previous one, advancing the capabilities of intelligent systems.

Rubrics for Calculation

Criteria	Weight age	Level 4 – Exemplary(90–100 Marks)	Level 3 – Proficient(75–89 Marks)	Level 2 – Developing(50–74 Marks)	Level 1 – Beginning(0–49 Marks)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand